

How AI Systems Propagate Semantic Failure in the Absence of a Correcting Model

Abstract

Prior analyses of AI-assisted workflow failure have assumed the presence of a human capable, in principle, of detecting when AI output has departed from the domain it is intended to represent. This assumption is not universal. A distinct and more dangerous failure mode arises when the user lacks a sufficiently explicit domain model to recognise drift, and the AI lacks persistent state across which drift could accumulate visibly. Under these conditions, each interaction appears locally correct. No signal of failure is produced. Confidence in the output is indistinguishable from confidence in correct output. Downstream systems, receiving artefacts that satisfy the shape of their expectations, propagate the drift further. The failure is not introduced by the AI. It is produced by the interaction between an AI that cannot detect incoherence and a human whose domain model was never sufficiently specified to provide a correcting force. This paper examines the structural conditions that produce this failure mode, the properties that make it self-concealing, and what would be required for it to be detectable before it becomes consequential.

1. The Assumed Correcting Force

The introduction of AI into knowledge work creates a transfer of activity from human actors to computational ones. That transfer is not neutral. When a human performs a task, they bring to it a model of the domain — however

incomplete, tacit, or inconsistently applied — against which output is continuously, if informally, checked. Errors that violate this model produce friction: hesitation, revision, escalation, or at minimum a vague unease that something is wrong. This friction is the system's primary self-correcting mechanism at the level of individual work.

Prior work on AI mapping failure has been premised on the assumption that this correcting force, while often insufficient, still operates. The practitioner who fails to decompose roles into functions will eventually encounter friction when role drift produces inconsistent outputs across sessions. The organisation that introduces AI into undefined workflows will eventually see the incoherence surface at integration boundaries. The outputs may be wrong for a long time, but the wrongness is eventually legible to someone.

This premise conceals an important boundary condition.

The correcting force is not a property of humans in general. It is a property of humans who have modelled their domain explicitly enough to recognise when output has departed from it. Where that modelling has not occurred, the correcting force does not exist. Friction is not generated. Drift is not noticed. The self-correcting mechanism is absent not because it failed, but because it was never constituted.

This is not a rare edge case. It is, for most users of AI in most organisational contexts, the normal condition.

2. The Structure of the Failure

The failure mode examined in this paper has the following structure.

A user whose domain model is underspecified engages an AI system to produce output within a domain they understand operationally but not foundationally. The AI produces output that satisfies the shape of the user's expectations — that is, output which corresponds to the user's model of what correct output looks like. Because the user's model is underspecified, the AI's underspecified output fits it without generating friction. The output is treated as authoritative.

That output is then passed to a downstream system: a process, a person, a tool, or another AI. The downstream system is similarly positioned — it expects input of a certain form, the input satisfies that form, and no mismatch is detected. The output of the downstream system is shaped by the input it received. Drift propagates.

Over time, the accumulated semantic distance between the output the AI is producing and the output the domain actually requires grows. But this distance is invisible at every layer. The user cannot see it because they lack the model against which it would be visible. The downstream systems cannot see it because they are consuming artefacts, not evaluating their correspondence to a domain. The AI cannot see it because it has no persistent state across interactions and no model of the domain independent of the context it is supplied.

There is no actor in the chain who is in a position to detect the failure. The failure therefore continues until it is forced into visibility by an external event — a regulatory examination, an audit, a consequential operational error — at which point its provenance is untraceable. No one knows how this happened because no record of the drift exists. Each individual interaction was locally

coherent. The failure lives only in the gap between them, and that gap was never observed.

3. Why the Output Appears Correct

It is important to be precise about why the output of an AI system in these conditions appears correct.

The AI is not malfunctioning. It is doing exactly what it is designed to do: producing output that is locally plausible within the context it has been supplied. If the context is underspecified, the output will be underspecified in corresponding ways. If the context contains implicit assumptions that diverge from the domain, the output will faithfully instantiate those assumptions. The plausibility of the output is not evidence of its correctness. It is evidence that the model is effective at pattern-matching within the supplied context.

The user's experience of the output as correct is also not irrational. Their assessment is made against the only model available to them — their own — and the output satisfies that model. If their model is incomplete, the fact that the output satisfies it tells them nothing about whether the output satisfies the domain. They have no independent access to the domain against which to check.

This is the central structural property of the failure: the signal that would indicate error — the experience of the output as wrong — requires a reference model that is precisely what is absent. The failure is not detectable with the resources available to the actors experiencing it.

4. The Confidence Inversion

A further property of this failure mode is that confidence and correctness are systematically decoupled, and in some conditions inversely related.

A practitioner with a well-developed domain model will experience friction when AI output departs from the domain. That friction is uncomfortable. It requires intervention, correction, and often a recognition that the context supplied to the AI was insufficient. It produces hesitation and qualification. The practitioner's confidence in the output is modulated by their capacity to detect its limits.

A practitioner without a well-developed domain model will experience no such friction. The output arrives complete, fluent, and formally appropriate. There is nothing to interrupt confidence. The practitioner's confidence in the output is, in the absence of a correcting model, a function of the output's surface properties: its fluency, its formal completeness, its correspondence to the expected shape of correct output.

This produces a confidence inversion. The practitioner least equipped to detect error is the practitioner most likely to treat the output as authoritative. High confidence in the output is therefore a more reliable indicator of structural risk than low confidence, in the class of cases where the user's domain model is underspecified.

This inversion has a direct implication for governance. Audit and review mechanisms that rely on human attestation — “has this been checked?” — provide no protection against this failure mode, because the check is performed by the same actor whose model is insufficient to detect the problem. The check is performed and passed. The record indicates review. The output

was never evaluated against a reference that could reveal its departure from the domain.

5. The Propagation Mechanism

Drift does not remain local. The architecture of modern knowledge work is composed of sequential transformations in which the output of one activity becomes the input of the next. Documents become requirements.

Requirements become designs. Designs become implementations.

Implementations become reports. Each transformation is constrained by what it receives.

In this architecture, a semantically drifted artefact does not merely introduce a local error. It reshapes the constraint space within which all subsequent transformations occur. Downstream systems adapt to what they receive. The implementation faithfully realises the design that faithfully realised the requirement that faithfully expressed the user's underspecified model of what a requirement should look like. At each stage, local coherence is maintained. At no stage is the growing distance from the domain visible.

The propagation is accelerated by two properties of AI systems in particular.

First, AI systems produce output at a rate that exceeds the rate at which domain errors would historically have been introduced by human actors performing the same transformations. The volume of semantically drifted material entering downstream systems is therefore higher than it would have been under equivalent human workflows.

Second, AI-produced output carries an implicit authority derived from its fluency and formal completeness. Outputs that read well, are structurally coherent, and satisfy formal expectations tend to receive less scrutiny than outputs that are visibly rough or provisional. The surface properties of AI output work against the scepticism that would be required to detect its semantic limits.

The combination produces a failure that is both faster and more deeply embedded than its historical analogues.

6. Dangerous Contexts

Not all domains are equally exposed. The failure is most consequential where four conditions coincide.

The first is high cost of semantic error. In domains where the meaning of an artefact has direct operational, financial, legal, or clinical consequences, the distance between what the output says and what the domain requires is not an abstraction. It produces harm that is proportional to how long the drift continued undetected.

The second is long feedback loops. Domains in which errors surface months or years after they are introduced provide the maximum opportunity for drift to propagate and become embedded before it is visible. Regulatory reporting, long-term financial modelling, clinical protocol documentation, and legal instrument drafting all exhibit this property.

The third is separation between the actor producing the output and the domain expertise required to evaluate it. Where the user is a specialist in the execution

of a task but not in the foundational semantics of the domain the task serves, the correcting force is structurally absent. A paralegal producing contract language within a framework they understand procedurally but not jurisprudentially. A financial analyst producing regulatory reports within a classification scheme they have never interrogated. A clinical administrator producing protocol documentation within a medical domain they navigate operationally.

The fourth is sequential transformation through multiple actors or systems, each of which receives the output of the previous stage as authoritative. Each additional transformation layer both extends the propagation and reduces the tractability of the original failure.

Where all four conditions are met, the uninstructed drift is not merely a quality problem. It is a systemic risk that is structurally invisible until it surfaces as a crisis.

7. What Detection Would Require

Detection of this failure mode cannot be achieved through the mechanisms that detect ordinary error, because it does not produce ordinary error signals. It produces plausible output. The question is what reference must exist for the departure from the domain to be visible.

The answer is precise: detection requires the existence, independent of the AI and independent of the user's model, of an explicit semantic contract that specifies what correct output in this domain looks like with sufficient precision to detect departure from it.

This contract must be prior to the interaction. It cannot be derived from the output, because the output is what is being evaluated. It cannot be derived from the user's model, because the user's model is what is being corrected. It must exist as an independent artefact that encodes the domain's requirements at a level of specificity sufficient to make the difference between correct and drifted output observable.

This is a demanding condition. In most organisational contexts, this contract does not exist in explicit form. Domain requirements are encoded in regulation, convention, and tacit expertise distributed across practitioners. The implicit assumption has been that human actors, drawing on this distributed knowledge, would provide the correcting force. AI systems make this assumption structurally untenable.

The second requirement for detection is that evaluation must occur against the contract, not against the output's surface properties. This rules out governance mechanisms that rely on attestation, peer review by actors without the relevant domain model, or automated consistency checking within the output's own framework. Each of these mechanisms can confirm that the output is internally coherent. None of them can confirm that the output corresponds to the domain.

The third requirement is that evaluation must occur before the output enters downstream transformation. Once the drift is embedded in artefacts that downstream systems have consumed and built upon, the cost of correction becomes proportional to the depth of the dependency, and the tractability of reversal decreases accordingly.

8. The Absence of Structural Remedy at the AI Layer

It is worth being explicit about what cannot address this failure mode.

Improved model capability does not resolve it. A more capable model will produce more fluent, more formally complete, and more contextually consistent output. If the context supplied is underspecified, a more capable model will produce a more convincing realisation of the underspecified input. Fluency is not correspondence to the domain. Greater fluency in the absence of a grounding contract increases the confidence inversion described in section four.

Prompt engineering does not resolve it. The problem is not that the user is asking the wrong question. The problem is that neither the user nor the system has access to the reference against which the answer's correctness could be evaluated. Better questions do not create this reference.

Explainability mechanisms do not resolve it. If the AI could explain how it arrived at its output, that explanation would be expressed in terms of the context supplied and the patterns applied to it. It would not constitute an evaluation of whether the output corresponds to the domain, because the AI has no independent model of the domain.

Audit logs do not resolve it. Logs establish what happened and in what sequence. They do not establish whether what happened corresponds to what the domain requires. A complete audit log of a process that propagated semantic drift throughout is a complete record of how the failure occurred, not a mechanism for preventing or detecting it.

The failure is not addressable at the AI layer because it is not produced at the AI layer. It is produced at the intersection of an underspecified human domain model and an AI system that operates faithfully within whatever model it is given.

9. The Reframing Required

This failure mode reframes the question that organisations must ask before introducing AI into consequential workflows.

The standard framing is: can the AI perform this task? This question is answerable, and the answer is increasingly often yes in a technical sense.

The reframing required is: does an explicit semantic contract exist against which the AI's output can be evaluated, and is that contract held by actors who are positioned to perform evaluation before output enters downstream transformation?

If the answer to this question is no, the introduction of AI does not introduce a new risk. It amplifies a pre-existing one. The human workflow was already operating without an adequate correcting mechanism. The AI removes whatever friction the inadequate mechanism was still providing, increases the volume and apparent authority of the output, and extends the propagation depth before the failure surfaces.

The absence of a semantic contract is not, in itself, a consequence of AI adoption. It is a prior condition that AI adoption makes consequential in new ways. Organisations that have operated for decades on the basis of tacit domain knowledge, informal coordination, and human repair have been running a risk

that remained manageable as long as the actors in the system could provide, however imperfectly, a correcting force. The withdrawal of that force — through AI adoption, through staff turnover, through scaling that exceeds the reach of tacit knowledge — reveals the risk as structural rather than operational.

The uninstructed drift is not a failure of AI systems. It is the first fully visible consequence of work that was never, in any meaningful sense, defined.

10. Conclusion

The failure mode described in this paper is structurally distinct from previously described AI workflow failures. It does not require misconfigured systems, inadequate prompts, or actors who ignore warning signals. It requires only that the user lacks the domain model to detect departure from the domain, and that the AI system operates faithfully within whatever model it is given.

Under these conditions, the output appears correct. Confidence is high. Downstream systems are satisfied. The failure propagates silently through successive transformations until an external event forces it into visibility. At that point, it is untraceable. No actor produced an observable error. The failure exists only in the accumulated distance between what was produced and what the domain required — a distance that no actor in the chain was positioned to measure.

Detection requires a semantic contract that is explicit, prior, independent of the AI, and held by actors positioned to evaluate output before it enters downstream transformation. Where this contract does not exist, AI adoption

does not introduce the risk. It removes the last friction that was making the risk tolerable.

The question is not whether AI systems can perform the work. The question is whether the work has been defined at a level that makes its performance evaluable. Until that question is answered, confidence in AI output is not evidence of correctness. It is evidence that the system has no mechanism for producing anything else.